

Introduction to the Internet of Things, Reference Architecture and Overview of Communication Technologies

Communication Technologies for the Internet of Things, Lesson 01

Catarina Silva

Outline (1/5)

Part I: Introduction to IoT and Reference Architecture

- Introduction to the lecturer and to the course
- The concept of Internet of Things and its historical evolution
- Prototyping platforms for IoT
- Reference architecture of the Internet of Things
- The machine to machine communication paradigm
- Differences between M2M and IoT

Outline (2/5)

Part I: Introduction to IoT and Reference Architecture (cont.)

- Analogue and digital quantities
- Analogue to digital and digital to analogue conversion

Outline (3/5)

Part II: Fundamentals of Communication Networks

- Internet architecture: edge, access networks and core
- Guided and unguided transmission media
- Packet switching, latency and packet loss
- The layered communication model and encapsulation

Outline (4/5)

Part II: Fundamentals of Communication Networks (cont.)

- Medium access: channel partitioning, taking turns and random access
- CSMA and CSMA/CA; the hidden terminal problem
- Mobile networks: the cellular concept, architecture and access scheduling
- Generations of mobile networks, mobility and 5G use cases
- Bridge to IoT: energy consumption, NB-IoT and LTE-M

Outline (5/5)

Part III: Overview of Wireless Communication Technologies

- The electromagnetic spectrum and the classification of networks by spatial scope
- IoT communication requirements compared with classical cellular networks
- Licensed and unlicensed spectrum
- Wired solutions: Ethernet and PowerLAN
- Short range wireless solutions: RFID, NFC, Bluetooth, ZigBee, Wi-Fi
- Technology selection criteria and design trade-offs

The Lecturer

- **Catarina Silva**, PhD in Informatics, from the Universities of Aveiro, Porto and Minho.
- Specialist in cybersecurity and privacy, with a research career grounded in communication networks and services.
- Participation in several nationally and European funded projects in the area of the Internet of Things.
- Lecturer responsible for this course.

Course Objectives

- Describe and apply the concepts required to define the architecture and design IoT applications.
- Select and apply the standards and communication technologies best suited to the requirements and constraints of a specific IoT solution.
- Select and apply the standards and data distribution technologies best suited to the scenario at hand.
- Design and implement IoT applications using communication solutions and message oriented middleware (MoM).

Course Syllabus (1/2)

- IoT reference architecture
- The machine to machine communication paradigm
- WAN access technologies, with emphasis on low power networks (LPWAN)
- Network technologies for IoT: IPv4, IPv6, 6LoWPAN, 6TiSCH, Neighbor Discovery, RPL, CoRPL, CARP

Course Syllabus (2/2)

- Session layer protocols for IoT: HTTP, CoAP, XMPP, AMQP, MQTT
- Message-Oriented Middlewares

Assessment

- Discrete assessment, split across two moments.
- Theoretical component: written exam taken in person, worth 50% of the final grade.
- Practical component: integrative mini-project, worth 50% of the final grade, delivered in the last lesson.
- In the Resit and Special assessment periods, assessment consists of a single exam covering both components.

Bibliography

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- Tsiatsis, V., Karnouskos, S., Höller, J., Boyle, D., Mulligan, C. (2014). *From Machine-to-Machine to the Internet of Things*. Academic Press.
- Scholz-Reiter, B., Michahelles, F. (2011). *Architecting the Internet of Things*. Springer.
- Minoli, D. (2013). *Building the Internet of Things with IPv6 and MIPv6*. Wiley.
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Integrative Mini-Project

- Development of a complete IoT application, from the sensor to the message broker.
- Use of at least one of the short range communication technologies covered in the course, such as NFC, BLE or Zigbee.
- Continuous development throughout the semester, with support in the practical lessons.
- Demonstration and delivery in the last lesson of the course.
- Detailed specification provided separately.

Part I

Introduction to IoT and Reference Architecture

What is the Internet of Things (1/2)

- The Internet of Things designates the set of physical objects equipped with sensors, processing capability and connectivity, able to exchange data with each other and with systems on the network without constant human intervention.
- The expression *Internet of Things* is generally attributed to Kevin Ashton, who used it in 1999 in the context of the MIT Auto-ID Center, associating it initially with radio frequency identification (RFID).

What is the Internet of Things (2/2)

- Cisco places the effective start of IoT (the moment when the number of devices connected to the Internet exceeded the number of people) between 2008 and 2009.
- The growth in the number of connected devices raises problems of scale in the collection, aggregation and analysis of data, a phenomenon associated with Big Data.

Prototyping Platforms: Arduino

- Arduino is an open source hardware and software platform, made up of single board microcontrollers and the corresponding development environment.
- It allows prototypes that read sensors and control actuators to be built quickly, without requiring digital electronics built from scratch.
- The hardware is distributed under a Creative Commons licence and the software under GNU LGPL or GPL licences, which allows any manufacturer to produce compatible boards.
- It is the reference platform for the practical lessons of this course.

Other Microcontroller Platforms (1/2)

- Teensy: small boards with processors more powerful than the classic Arduino, useful when more computing capability is needed in little space.
- MSP430: family of Texas Instruments microcontrollers aimed at applications with very low energy consumption.
- NodeMCU: based on the ESP8266 module, it integrates Wi-Fi into the microcontroller itself, which simplifies IoT solutions connected directly to the local network.

Other Microcontroller Platforms (2/2)

- STM32: family of 32 bit microcontrollers from STMicroelectronics, used in applications requiring more performance and more peripherals.
- PocketBeagle: board from the BeagleBoard family, with a processor closer to a computer than to a simple microcontroller.

Single Board Computers: Raspberry Pi (1/2)

- The Raspberry Pi is a family of single board computers developed in the United Kingdom by the Raspberry Pi Foundation, initially to promote the teaching of computing in schools.
- Unlike a microcontroller, it runs a complete operating system, which makes it easier to develop more complex applications and to integrate with IP networks.

Single Board Computers: Raspberry Pi (2/2)

- In this course, the Raspberry Pi typically takes the role of gateway between short range devices and the local network or the Internet.
- Alternatives exist on the market, such as the ASUS Tinker Board, the NVIDIA Jetson Nano, the Onion Omega2+, the BeagleBone Black and the Orange Pi, with different balances between price, performance and consumption.

Internet of Things Reference Model (I) (1/2)

- The IoT World Forum Reference Model, proposed by Cisco in 2014, organises an IoT solution into seven layers, from the physical capture of data to its use by business processes.
- Layer 1, Physical Devices and Controllers: the sensors, the actuators and the other objects that are the origin and the final destination of the information.
- Layer 2, Connectivity: the networks and communication protocols that carry data between devices, gateways and the network.

Internet of Things Reference Model (I) (2/2)

- Layer 3, Edge Computing (Edge/Fog Computing): processing near the origin of the data, which filters, converts or reacts to events before sending them to the network.
- Layer 4, Data Accumulation: storage of the collected data, turning event streams into persistent data sets.

Internet of Things Reference Model (II) (1/2)

- Layer 5, Data Abstraction: aggregation and normalisation of data coming from multiple sources, making it accessible to applications in a consistent way.
- Layer 6, Application: interpretation of the data for concrete purposes, such as reporting, analysis or automatic process control.

Internet of Things Reference Model (II) (2/2)

- Layer 7, Collaboration and Processes: integration of the information into business processes and into decision support for people and organisations.
- This model serves as a reference throughout the course: layers 1 and 2 correspond to the communication technologies studied in this and in the coming lessons, and the upper layers relate to the middleware and data distribution protocols covered later.

The Machine-to-Machine Communication Paradigm (1/2)

- Machine-to-Machine (M2M) designates the interconnection of machines or devices for the purpose of remote monitoring, control and data exchange, without direct human intervention.
- An M2M area network is made up of nodes with embedded sensing, actuation and communication modules.

The Machine-to-Machine Communication Paradigm (2/2)

- Protocols used in these local networks include ZigBee, Bluetooth, Modbus, M-Bus, Wireless M-Bus, Power Line Communication (PLC), 6LoWPAN and IEEE 802.15.4.
- The network that connects the different M2M area networks to each other is typically IP based, whether over wired or wireless media.

M2M Gateways

- M2M area networks often use proprietary or non IP protocols, which prevents direct communication between nodes of distinct M2M networks.
- The M2M gateway solves this limitation by translating between the local protocol of the M2M area network and the IP protocol used in the communication network.
- In this course, the gateway role will be taken by a Raspberry Pi in several practical lessons.

Differences between IoT and M2M (I)

- Communication protocols: M2M relies mostly on proprietary or non IP protocols inside the local area network; IoT may also use non IP protocols in sensor networks, but tends to expose the data through IP protocols on the wider network.
- Homogeneity: M2M networks typically have homogeneous machine types within the same area network. In IoT, the term “thing” refers to any physical object with a unique identifier, able to sense and communicate with the environment and with other applications, which results in greater heterogeneity.

Differences between IoT and M2M (II)

- Technological emphasis: M2M rests mostly on embedded hardware modules; IoT gives more weight to software, to data integration and to analysis.
- Data collection and analysis: in M2M, data is typically collected and stored in local infrastructure, tied to point solutions; in IoT, data is collected and processed mostly in the cloud, whether public, private or hybrid.
- Applications: M2M applications are normally local, such as diagnostics or service management; IoT applications reside in the cloud and include advanced analytics, remote diagnostics and enterprise integration.

The “Things”: Sensors (1/2)

- The term “thing” represents, in abstract terms, any physical element able to produce or consume information in an IoT solution, typically a sensor or an actuator.
- A sensor may be simple, dedicated to monitoring a single physical quantity, such as temperature or humidity.

The “Things”: Sensors (2/2)

- A sensor may also be complex, combining several quantities to infer behaviour, for instance the health state of a person or an animal, or patterns of social and financial activity.
- The choice between simple and complex sensors directly constrains the communication and processing requirements of the solution.

Analogue to Digital and Digital to Analogue Conversion

- Microprocessors are digital machines, operating on discrete values.
- The physical world observed by sensors is essentially analogue, with quantities that vary continuously in time.
- An analogue to digital converter (ADC) transforms the continuous signal coming from the sensor into a numeric sequence that the processor can handle.
- When the digital system needs to act on the physical world, the reverse process is required, carried out by a digital to analogue converter (DAC).

Sampling and the Nyquist Theorem (1/2)

- Sampling consists of measuring the input signal periodically, producing a sequence of values that represents it over time.
- If the sampling frequency is too low, that sequence does not allow the original signal to be correctly reconstructed.

Sampling and the Nyquist Theorem (2/2)

- The Nyquist Theorem, formulated by Harry Nyquist, states that the sampling frequency must be at least twice the highest spectral component present in the input signal: $f_s = 2B$.
- In practice, a low pass filter is used to bound B , and a sampling frequency five to ten times higher than the highest spectral component of the signal is adopted, to provide a safety margin.

Quantisation Error (1/2)

- When converting the amplitude of an analogue signal into a digital value, precision is lost, because the number of available discrete values is finite.
- This loss of precision is called quantisation error and depends directly on the number of bits used in the conversion.
- With 3 bits of resolution only 8 distinct levels are obtained, which produces a coarse representation of the original signal.

Quantisation Error (2/2)

- With 4 bits of resolution 16 distinct levels are obtained, already a noticeably better approximation.
- The larger the number of bits, the smaller the quantisation error, but also the larger the volume of data produced.

Sampling Rate and Resolution: Practical Examples (1/2)

- Classical telephony: sampling at 8 kHz, with 8 bits per sample, gives a transmission rate of 64 kbit/s; one minute of conversation takes about 469 KB.
- Audio CD: sampling at 44.1 kHz, with 16 bits per sample and two channels, gives a rate of about 1411 kbit/s; one minute of music takes about 10.5 MB, and a 74 minute CD holds about 740 MB.
- Other common sampling rates: 48 kHz for DAT, DV and DVD-Video; 96 kHz for DVD-Audio.

Sampling Rate and Resolution: Practical Examples (2/2)

- In terms of resolution: low quality applications use 8 bits, CD and DV use 16 bits, DVD uses 24 bits, and the internal representation in software often uses 32 bit floating point.
- These parameters illustrate a central trade-off in the design of any data acquisition system in IoT: quality of the reconstructed signal against the volume of data to transmit or store.

Summary of Part I

- The Internet of Things results from the convergence of instrumented physical devices, communication networks and data processing systems.
- The Cisco reference model organises this convergence into seven layers, from the physical device up to integration into business processes.
- The M2M paradigm predates IoT and shares many concepts with it, but is distinguished by lower heterogeneity and greater reliance on proprietary protocols.
- Before any data reaches the network, it has to be captured from the analogue world and converted to the digital domain, a process that constrains the quality and the volume of the information available to the layers above.

Part II

Fundamentals of Communication Networks

Why Study Networks in an IoT Course

- An IoT solution is, first of all, a communication network: without data transport, a sensor is just an isolated measuring instrument.
- Layers 1 and 2 of the reference model seen in Part I correspond exactly to the topics of this part: the physical medium and the way several devices share it.
- Practically every technology studied in the rest of the semester (BLE, ZigBee, LoRa, NB-IoT, Wi-Fi) is a variation on the same set of fundamental problems.
- Understanding those problems in the abstract makes it possible to compare technologies instead of memorising them one by one.

Architecture of a Communication Network

- **Edge:** the hosts that run applications and that originate or consume the information.
- **Access networks:** the links, wired and wireless, joining those devices to the first equipment of the network.
- **Core:** the mesh of interconnections between networks, typically IP based.
- In an IoT solution, sensors and actuators occupy the edge, the short range wireless technology is the access network, and the gateway is the point of passage to the core. It is in the access network that range, consumption and cost per device are decided.

Protocols and Standardisation

- A protocol is a set of rules defining the order and format of the messages exchanged, as well as the actions to take upon receiving each message.
- Without a common protocol, two pieces of equipment connected to the same physical medium remain unable to communicate: the physical link is a necessary condition, but not a sufficient one.
- Standards ensure that implementations from different manufacturers interoperate, which is decisive in a market as fragmented as that of IoT.
- Bodies relevant to this course: IETF (RFCs), IEEE (802.11, 802.15.4), 3GPP (cellular networks) and industry alliances such as the Bluetooth SIG, the Connectivity Standards Alliance and the LoRa Alliance.

Transmission Media: Guided and Unguided

- Transmitting data between two points requires, first, converting that data into physical signals the medium can carry.
- In **guided media** (copper, optical fibre, coaxial cable), signals propagate along a fixed physical support. They offer predictability and immunity to interference, but impose fixed infrastructure, incompatible with mobile devices or with dispersed installations.
- In **unguided media**, signals propagate freely in space, in the form of radio or optical waves.
- This distinction is the source of nearly all the differences in behaviour between the technologies studied in this course.

The Shared Medium and its Consequences

- The decisive consequence of an unguided medium is that it becomes **shared**: everything transmitted on a given frequency in a given area is received by all devices within range.
- From this follows **half-duplex** operation: a device does not transmit and receive at the same time on the same frequency, because its own signal dominates the receiver.
- The medium also introduces effects that guided media do not have: reflections off surfaces, obstruction by obstacles and noise from other sources.
- The whole of Part III of this lesson deals with unguided technologies: these limitations are the common denominator of all of them.

Packet Switching, Delay and Loss

- Messages are not transmitted whole: they are divided into packets, each with a maximum size and with the information needed to determine its destination.
- **Store and forward:** the packet must arrive in full at each piece of equipment before being retransmitted, which makes the delay accumulate at every hop. In an IoT mesh network such as ZigBee, this effect determines the end to end delay.
- When traffic exceeds the capacity of the outgoing link, packets wait in queues; if the queue fills up, they are discarded.
- In IoT, the bottleneck is typically at the gateway, where the aggregate traffic of dozens of sensors has to fit into a single outgoing link.

Latency: Where it Comes From and Why it Matters

- Latency breaks down into four components: processing at the equipment, queuing, transmission and propagation.
- **Transmission delay:** time to place all the bits of the packet on the link, given by L/R , where L is the packet size and R the link speed. It depends on the throughput of the technology.
- **Propagation delay:** time the signal takes to physically travel the link, given by d/s . It depends only on distance and on the physics of the medium, and does not shrink by raising the throughput.
- This last component will reappear later: it is what produces the vulnerability period that makes collisions inevitable, even in protocols that listen to the channel before transmitting.

Layered Organisation: Principle

- Protocols are organised into layers, each solving a well delimited part of the overall communication problem.
- Each layer offers a service to the layer above and uses the services of the layer below, without knowing its internal workings. This allows the implementation of one layer to be replaced without changing the others.
- It is thanks to this principle that the same IoT application runs over Wi-Fi, over Ethernet or over a cellular network, with no change to the code.
- It is also why it makes sense to study medium access in the abstract: the conclusions apply to any technology occupying that layer.

The TCP/IP Model (1/2)

- **Application layer:** application oriented services. In IoT, this is where MQTT, CoAP and AMQP live, studied in Lessons 11 and 12.
- **Transport layer:** identifies the applications on each device and handles segmentation, flow control and retransmission. The protocols are TCP and UDP.
- **Network layer:** identifies devices and forwards packets, with fragmentation where needed. The protocol is IP.

The TCP/IP Model (2/2)

- **Link layer:** encapsulates data into frames, detects errors and handles transmission between directly connected devices. It is the layer of Wi-Fi and IEEE 802.15.4.
- **Physical layer:** converts bits into radio, electrical or optical signals.
- The two lower layers are the central object of this course, and are where IoT technologies depart most from conventional solutions.
- The later OSI model breaks the application layer into application, presentation and session. It is from that model that the designation “session layer protocols” comes, which gives its name to Lesson 11.

Encapsulation (1/2)

- At each layer, the data received from above is wrapped in a header of its own before being handed to the layer below.
- The application message becomes a **segment** at the transport layer, a **datagram** at the network layer and a **frame** at the link layer.
- At the destination, the reverse process removes each header in turn, until the original message is delivered to the application.

Encapsulation (2/2)

- Every header added represents data that occupies space on the medium and energy to transmit, without carrying any useful application information.
- In conventional networks this cost is negligible. In IoT, where the useful information may be a single temperature value of a few bytes, headers can exceed the content itself.
- This observation is what motivates 6LoWPAN, studied in Lesson 09, whose essential function is to compress IPv6 headers down to sizes compatible with the frames of a sensor network.

The Data Link Layer

- The link layer establishes the communication channel between adjacent nodes, a node being any end device, router or switch.
- Its responsibility is to transfer frames, which contain the network layer datagrams, between directly connected nodes.
- It is the only layer whose protocol may change at every hop: a datagram can cross Wi-Fi, then Ethernet, then fibre, without ever being altered.
- It is also the layer where the central problem of this part of the lesson lives: access to the transmission medium.

Link Types: Point to Point and Shared

- **Point to point links:** terminated directly between two devices. They operate in full-duplex and require no coordination.
- **Shared links with broadcast:** they connect multiple devices to the same medium, as in Wi-Fi or IEEE 802.15.4. They operate in half-duplex.
- Shared links necessarily require a coordination protocol, called Medium Access Control (MAC).
- Practically every wireless IoT technology is a shared link: for them, medium access control is not an optional detail.

Link Layer Services

- **Framing:** adds the link layer addresses and the fields needed for additional functionality.
- **Access to the transmission medium:** defines the rules for when a frame may be sent. It may be absent on point to point links in full-duplex.
- **Reliable delivery between adjacent nodes:** used on links with many errors. In Wi-Fi and IEEE 802.15.4, frames are acknowledged by the receiver.
- **Error detection and correction:** identifies frames corrupted by the medium, typically through a check code.

The Medium Access Problem (1/2)

- On a shared medium, if two devices transmit at the same time, the signals overlap and interfere with each other.
- The result is a **collision**: neither transmission is correctly received, and both are wasted.
- What makes it worse is that, on a radio medium, the collision happens at the **receiver**, and is therefore unknown to the transmitters at the moment it occurs.

The Medium Access Problem (2/2)

- A medium access protocol is therefore a **distributed algorithm**: each device decides when to transmit using only local information.
- A good protocol should approach the full channel throughput when only one node transmits, share capacity fairly when several transmit, and work without central coordination.
- In IoT a further criterion applies, one that conventional networks ignore: every collision is energy spent with no result at all, and energy is the scarcest resource on the device.

Taxonomy of Multiple Access Protocols

- **Channel partitioning:** the channel is divided into portions (of time, of frequency or of code) and each device uses its own exclusively.
- **Taking turns:** devices transmit in turn, coordinated by polling or by the circulation of a token.
- **Random access:** each device transmits when it has data, accepting the risk of collision and resolving it when it occurs. This is the ALOHA and CSMA family.
- None of these families is universally superior: the choice depends on the traffic load, the number of nodes and the tolerance to delay.

Channel Partitioning: TDMA

- In Time Division Multiple Access, time is divided into frames, and each frame into slots assigned to specific devices.
- Each device transmits only in its own slot, at the full speed of the channel, and stays silent during the others.
- **Advantages:** complete absence of collisions, deterministic maximum delay and the possibility of switching the radio off outside its own slot, with a direct energy gain.
- **Disadvantages:** a device with no data wastes its slot, and strict time synchronisation between all nodes is required.

Channel Partitioning: FDMA, CDMA and OFDMA

- **FDMA:** the band is divided into sub-bands and each device transmits continuously in its own. There are no collisions, but each device is limited to a fraction of the total throughput, even when the others are idle.
- **CDMA:** all devices transmit in the same band at the same time, separated by orthogonal codes. It is robust to narrowband interference, but requires tight power control.
- **OFDMA:** divides the band into many narrow orthogonal subcarriers, assigning each device a set of subcarriers during a set of intervals. It combines the logic of FDMA and TDMA and is the basis of 4G, 5G and Wi-Fi 6.

Taking Turns: Polling and Token

- In **polling**, a central node asks each device in turn whether it has data to transmit. It removes collisions and allows very simple peripheral devices, but introduces the delay of the polling itself and creates a single point of failure.
- This is exactly the model of classic Bluetooth, in which the master device polls the slaves of the piconet, as will be seen in Lesson 04.
- In **token passing**, a frame circulates among the devices granting the right to transmit to whichever holds it. It requires no central node, but losing the token paralyzes the network.
- Both are efficient under high load and inefficient under low and sporadic load, which is precisely the typical profile of IoT traffic.

Random Access: ALOHA

- In pure ALOHA, a device transmits as soon as it has data, with no prior check of the state of the channel.
- If it receives no acknowledgement, it assumes a collision occurred and retransmits after a random interval, precisely to avoid repeating the collision.
- It is the simplest protocol possible and the one that allows the cheapest device, but its maximum efficiency is low, around 18% of channel capacity.
- Slotted ALOHA doubles that efficiency by forcing transmissions to begin at predefined instants; this is the access principle of LoRaWAN and Sigfox.

CSMA: Listen Before Transmitting (1/2)

- Carrier Sense Multiple Access adds one simple step to random access: listen to the channel before transmitting.
- If the channel is busy, the device defers transmission; if it is free, it transmits.
- This single change drastically reduces collisions compared with ALOHA, and is therefore the basis of the vast majority of random access protocols in use.

CSMA: Listen Before Transmitting (2/2)

- Collisions do not disappear, however, because of the **propagation delay**: two nodes may find the channel free at practically the same moment and both transmit.
- The interval during which this is possible is called the vulnerability period, and grows with the distance between the nodes.
- An important result follows: CSMA degrades as the network grows in physical extent, and not only in number of nodes.

CSMA/CA: Collision Avoidance (1/2)

- In wired networks there is the CSMA/CD variant, in which the device listens to the medium while transmitting and aborts the transmission as soon as it detects a collision.
- On a radio medium that detection is impractical: the transmitter cannot listen to the medium while transmitting, because its own signal saturates the receiver.
- CSMA/CA therefore replaces detection with **prevention**: even when it finds the channel free, the device waits a random interval before transmitting, desynchronising nodes that were waiting for the same moment.

CSMA/CA: Collision Avoidance (2/2)

- Since the sender cannot tell by itself whether the frame arrived, the receiver must **acknowledge** it explicitly; the absence of an acknowledgement is read as a collision.
- On failure, the random waiting interval is doubled at each new attempt, so as to relieve the channel when load is high.
- It is the protocol of Wi-Fi and of IEEE 802.15.4, the standard underlying ZigBee and 6LoWPAN, and will therefore reappear in several lessons of this course.

The Hidden Terminal Problem

- Consider three nodes in a line: A and C are both within range of B, but too far apart to hear each other.
- A listens to the channel, finds it free (because it does not hear C) and transmits to B; C does the same, at the same time. The two transmissions collide at B, and neither A nor C has any way of knowing.
- Channel listening fails here because the decision is taken with local information, at the sender, while the collision happens at the receiver.
- The classic solution is a prior exchange of short reservation messages: because the authorisation comes from the **receiver**, it also reaches the nodes hidden from the sender, which thus learn that they must stay silent.

Comparing Medium Access Protocols

- Under **high load**, channel partitioning and taking turns are superior: they waste no capacity on collisions and guarantee bounded delay.
- Under **low and sporadic load**, random access is superior: an isolated node uses the whole channel immediately, without waiting for its turn or its slot.
- IoT traffic is typically sporadic and small, which explains the predominance of CSMA and ALOHA variants in short range and unlicensed long range technologies.
- Cellular networks, which have to serve human and machine traffic simultaneously with service guarantees, adopt the opposite logic, of centrally scheduled access.

Addressing and Wireless Local Network Architecture

- The link layer address, or MAC address, identifies devices connected to the same network and is set by the manufacturer of the interface. On local networks it typically has 48 bits.
- In sensor networks its size matters: IEEE 802.15.4 admits short 16 bit addresses precisely to reduce the weight of the header in very small frames.
- In a wireless local network, each station (STA) associates with an access point (AP), forming a BSS (Basic Service Set). Access points periodically announce the network identifier (SSID).
- Medium access is done by CSMA/CA, with acknowledgement of every frame, and the frame includes duration fields to manage the reservation of the medium.

Mobile Networks: the Problem to Solve

- The technologies seen so far cover at most a few tens or hundreds of metres, and assume practically static devices.
- Mobile networks solve two further problems: covering entire territories with limited spectrum, and keeping communication alive while the device moves.
- The answer to both is the same structural idea, which gives the technology its name: dividing the territory into **cells**.
- It is this architecture that NB-IoT and LTE-M reuse to give connectivity to IoT devices scattered across an entire region.

The Cellular Concept

- Instead of a single high power transmitter, the territory is divided into cells, each served by a base station of moderate power.
- Because the signal attenuates with distance, the same frequency can be **reused** in another cell far enough away. This is what makes it possible to serve millions of devices with limited spectrum.
- Total system capacity therefore grows by reducing cell size, and not by increasing transmission power.
- Smaller cells mean shorter distance to the device, hence lower power demanded of the terminal, which has a direct energy impact on IoT devices.

Architecture of a Cellular Network

- **User equipment (UE):** the terminal, which in IoT is typically a sensor with an integrated cellular module.
- **Radio access network (RAN):** the set of base stations, responsible for the radio interface and for access scheduling.
- **Core network:** authentication, mobility management, billing and connection to the Internet.
- The subscriber identity resides in the SIM card, today often in the form of a soldered eSIM, better suited to sealed IoT devices installed in the field. Network management belongs to the operator, which brings service guarantees but also a recurring cost per device.

Medium Access in Cellular Networks (1/2)

- Unlike Wi-Fi, the cellular network does not let terminals contend for the medium: access is **centrally scheduled** by the base station.
- The base station knows the demand and the channel quality of each terminal, and distributes time and frequency resources among them.
- In the downlink, the base station is the only transmitter, so there are no collisions; in the uplink, each terminal transmits only when and where it is told to.

Medium Access in Cellular Networks (2/2)

- This model allows guarantees that random access cannot offer: bounded latency, reserved throughput and differentiated priorities. It also allows tight power control, which saves battery.
- The price is complexity: the synchronisation and control signalling make the terminal more expensive and more power hungry than a simple random access radio.
- It is exactly this trade-off that separates cellular solutions from unlicensed LPWAN, compared at the end of this part.

The Random Access Channel

- One problem remains: a terminal that has no resources assigned yet needs to request them, but has no scheduled means of doing so.
- The solution is to reserve a random access channel, in which terminals contend by sending a preamble chosen from a limited set.
- If two terminals choose the same preamble at the same instant, a collision occurs and both repeat the procedure after a random interval.
- A cellular network therefore combines the two access families: random for the initial request, scheduled for the traffic that follows. In massive IoT, with tens of thousands of devices waking up at once, it is precisely this channel that becomes the bottleneck.

Generations and Multiple Access Techniques

- **1G:** analogue, FDMA. **2G (GSM):** digital, FDMA combined with TDMA. **3G (UMTS):** CDMA.
- **4G (LTE):** OFDMA, with a lower peak power variant in the uplink, to spare the terminal battery. **5G (NR):** OFDMA with configurable subcarrier spacing.
- The reading of this evolution that matters for IoT: each generation sought finer granularity in resource assignment, to accommodate increasingly heterogeneous traffic, machine traffic among it.
- Only from 4G onwards, with NB-IoT and LTE-M introduced in 2016, did cellular standardisation begin to treat IoT as a first class use case.

Mobility: Handover and Roaming

- As the terminal moves, the signal of the current cell degrades and that of a neighbouring cell improves, making it necessary to transfer the connection, a process called **handover**.
- The network also keeps a record of the approximate location of each terminal, so that it can reach it when there is traffic addressed to it.
- **Roaming** extends this mechanism to networks of other operators, which often justifies SIMs with permanent roaming in IoT equipment deployed internationally.
- Many IoT devices are fixed and need no handover: NB-IoT therefore does without it in active connection, while LTE-M keeps it to serve cases with real mobility.

5G: eMBB, URLLC and mMTC

- **eMBB:** high throughput mobile broadband, the natural continuation of previous generations.
- **URLLC:** very low latency and very high reliability, for industrial automation, autonomous vehicles and critical applications.
- **mMTC:** massive machine communication, sized for densities of the order of one million devices per square kilometre. This is the category relevant to IoT, and it is where the evolution of NB-IoT and LTE-M integrated into 5G belongs.
- The three families demand mutually contradictory optimisations. The 5G answer is **network slicing**: several logical networks over the same physical infrastructure, each sized for one traffic profile.

From Network Theory to IoT Technologies

- Everything seen in this part describes networks designed for people: high throughput, guaranteed power supply and tolerance for complex protocols.
- IoT technologies solve the same problems (medium access, addressing, encapsulation, routing) under radically different constraints.
- Each of the coming lessons can therefore be read as a concrete answer to these general problems, with a different trade-off between range, consumption, throughput and cost.
- The next slides make that bridge explicit, before we move on to the overview of technologies.

The Layered Model in an IoT Solution (1/2)

- **Physical and link layers:** IEEE 802.15.4, Bluetooth Low Energy, LoRa, NB-IoT. This is where range and consumption are decided, and where the medium access control studied in this part lives.
- **Network layer:** IPv6 compressed by 6LoWPAN, with mesh routing by RPL. Topics of Lessons 08, 09 and 10.
- **Transport layer:** predominantly UDP rather than TCP, being lighter and requiring no connection establishment, which suits devices that wake up, send and go back to sleep.

The Layered Model in an IoT Solution (2/2)

- **Session and application layers:** CoAP, MQTT and AMQP, designed for small headers and for the publish and subscribe pattern. Topics of Lessons 11 and 12.
- The gateway is the point where the lower layers change: below it, a short range technology; above it, conventional IP over the access network of the building or of the operator.
- This translation is exactly the function of the M2M gateway described in Part I, now with the technical vocabulary needed to describe it precisely.

Medium Access and Energy Consumption (1/2)

- On a battery powered device, the radio is typically the component that consumes most energy, and it consumes it both transmitting and listening.
- Listening to the channel is therefore not free: CSMA forces the device to keep the radio on before every transmission, and idle listening is one of the main sources of waste.
- Every collision means energy spent on a lost transmission, on top of which comes the energy of the subsequent retransmission.

Medium Access and Energy Consumption (2/2)

- From this follows the renewed interest in TDMA for IoT: knowing its slot in advance lets the device switch the radio off completely for the rest of the time.
- That is precisely the principle of 6TiSCH, studied in Lesson 10, which combines time slots with frequency hopping over IEEE 802.15.4.
- The determining parameter is the **duty cycle**: the fraction of time the radio is active. Lifetimes of several years are only possible with duty cycles well below 1%.
- The design consequence is clear: in IoT, the medium access protocol is not chosen for the throughput it offers, but for the time it lets the radio stay off.

Cellular Networks for IoT: NB-IoT and LTE-M (1/2)

- **NB-IoT** is a very narrowband technology, occupying about 180 kHz, which can be deployed inside an LTE carrier, in its guard band or standalone.
- It offers throughput of the order of a few tens of kbit/s and a coverage improvement of about 20 dB over earlier cellular technologies, which lets it reach basements and deep indoor areas.
- It deliberately gives up mobility in active connection and voice, in order to simplify the terminal and reduce consumption and cost.

Cellular Networks for IoT: NB-IoT and LTE-M (2/2)

- **LTE-M**, also called eMTC, occupies about 1.4 MHz and offers throughput of the order of Mbit/s, supporting mobility with handover and voice over IP.
- Both rely on two mechanisms that are decisive for battery life: a power saving mode in which the device stays registered but unreachable for long periods, and extended discontinuous reception, which spaces out the listening instants.
- These mechanisms are the transposition, into the cellular world, of the reduced duty cycle principle: the device stays on the network without keeping the radio on.
- The choice between the two comes down to mobility and throughput (LTE-M) against coverage, cost and battery life (NB-IoT).

Licensed Cellular and Unlicensed LPWAN

- Cellular solutions use licensed spectrum and scheduled access, with service guarantees and operator infrastructure, but with a recurring cost per device.
- Unlicensed LPWAN, such as LoRa and Sigfox, use shared spectrum and random access, without guarantees, but allow owned infrastructure and need no subscription.
- The essential difference lies in medium access: centralised scheduling on one side, pure contention on the other, with direct consequences for latency and predictability.
- Neither is universally better: the choice depends on the application, on device density and on the business model. A topic revisited in Lesson 03.

Summary of Part II

- Layered organisation lets the same application run over any access technology, but each layer adds headers that, in IoT, may exceed the useful information.
- On shared media, the central problem is medium access: channel partitioning, taking turns or random access. Random access suits sporadic traffic and underpins Wi-Fi, IEEE 802.15.4 and LPWAN; scheduled access underpins cellular networks and their guarantees.
- Mobile networks solve coverage and mobility through the cellular concept, and reserve random access only for the initial resource request.
- In IoT, the dominant criterion when choosing a medium access protocol is not throughput, but the fraction of time it allows the radio to stay off.

Part III

Overview of Wireless Communication Technologies

Classification of Networks by Spatial Scope (1/2)

- Computer networks are classified by the physical extent they cover, and that extent directly constrains which communication technology is applicable.
- At the smallest scale are near field networks (NFC) and body area networks (BAN), with ranges from centimetres to a few metres.
- Next come personal (PAN), home (HAN) and local networks (LAN and WLAN), typical of a building or a dwelling.

Classification of Networks by Spatial Scope (2/2)

- At larger scales are campus (CAN), metropolitan (MAN) and wide area networks (WAN), up to the Internet itself.
- An IoT solution usually combines several of these scales: sensors connected by a PAN, aggregated at a gateway, which communicates with the cloud over a WAN.

Characteristics of IoT Communications (1/2)

- IoT communications differ from conventional cellular communications through a set of requirements of their own.
- Low cost per device and per connection, since the number of nodes is high.
- Low energy consumption and long battery life, often measured in years.
- Very high number of simultaneous connections per cell or per gateway.

Characteristics of IoT Communications (2/2)

- Variable throughput requirements, but almost always modest compared with those of human communications.
- Small devices, with limited processing and storage capability, which calls for simple architectures and protocols.

Diversity of IoT Application Requirements (1/2)

- No single communication technology serves every IoT application: requirements vary significantly from scenario to scenario.
- The bandwidth needed depends on the volume of information to transmit, which differs between a temperature sensor and a surveillance camera.
- The range required separates short range applications, confined to a building, from long range applications, spread across a region.
- Energy autonomy is decisive in battery powered devices without frequent maintenance.

Diversity of IoT Application Requirements (2/2)

- Quality of service requirements vary with the criticality of the application.
- This diversity is what justifies making the radio spectrum available in several frequency bands, suited to different use cases.

General Connectivity Architecture in IoT (1/2)

- At the base of the architecture are the sensors and actuators, connected by short range wireless technologies or by wired media.
- These devices concentrate at gateways, industrial or domestic, which aggregate the traffic of the local sensor network.
- The gateway connects to the transport infrastructure, which may use cellular networks, optical fibre, cable or satellite links.

General Connectivity Architecture in IoT (2/2)

- At the top are the cloud storage services and the analytics and intelligence layers that process the collected data.
- Each segment of this chain imposes distinct requirements, and it is that difference which explains the coexistence of several technologies in a single solution.

Overview of Technical Solutions (1/2)

- **Cellular M2M solutions:** NB-IoT, eMTC and LTE V2X, operated in licensed spectrum by mobile operators.
- **Non cellular long range solutions:** LoRa, Sigfox, Ingenu and the Weightless family (N, P and W), in unlicensed spectrum.
- **Short range solutions:** Bluetooth and BLE, ZigBee, Z-Wave, Wi-Fi, NFC and RFID.

Overview of Technical Solutions (2/2)

- **IoT oriented Wi-Fi variants:** IEEE 802.11ah, designed for greater range and lower consumption, and IEEE 802.11p, aimed at vehicular communications.
- **Wired solutions:** Ethernet and Powerline, applicable where fixed infrastructure is available.

Licensed and Unlicensed Spectrum (1/2)

- **Licensed spectrum:** assigned to operators under licence, it offers better interference management, greater network security and higher reliability, with quality of service guarantees. It allows the existing cellular infrastructure and device ecosystem to be reused.
- **Unlicensed spectrum:** it requires no licence fees, which lowers the cost of operation, but is subject to regulatory limits on radiated power (EIRP restrictions) and offers no guaranteed quality of service.
- Any device may access unlicensed spectrum, provided it meets the technical conditions set in the applicable regulation.

Licensed and Unlicensed Spectrum (2/2)

- Short range and delay tolerant applications are the typical use cases for unlicensed spectrum.
- The choice between the two regimes is, in practice, a trade-off between cost of operation and service guarantees.

Wired Solutions: Ethernet (IEEE 802.3) (1/2)

- A low cost, highly reliable technology, with bandwidth performance far above that of the wireless alternatives.
- It allows devices to be powered through the cable itself, using Power over Ethernet, given additional circuitry and components.
- It requires a permanent cable connection, which restricts mobility and makes installation expensive in already constructed environments.

Wired Solutions: Ethernet (IEEE 802.3) (2/2)

- It draws high and continuous current, typically of the order of 200 to 300 mA, incompatible with battery powered devices.
- The connector is bulky, which limits its application in small devices.

Wired Solutions: PowerLAN (1/2)

- It consists of carrying Ethernet frames over the existing electrical network, taking advantage of the cabling already installed in the building.
- It should not be confused with Power over Ethernet: in that case, the data network carries power; in PowerLAN, the electrical network carries data.
- There is a wide diversity of models and characteristics on the market, some integrated with Wi-Fi so as to act as coverage extenders.

Wired Solutions: PowerLAN (2/2)

- Advertised speeds can reach 2 Gbit/s, but real performance depends heavily on distance and on the quality of the electrical installation.
- Electrical noise introduced by other equipment connected to the same network can compromise the operation of the link.

RFID: Radio Frequency Identification (1/2)

- A technology that uses radio waves to read information stored in a tag attached to an object, originating in work from the 1940s, with the first patent granted to Charles Walton in 1983.
- It allows objects to be identified, metadata to be recorded or a target to be controlled individually, without requiring direct line of sight between reader and tag.
- An RFID system is made up of a tag, a reader (also called interrogator) and an antenna.

RFID: Radio Frequency Identification (2/2)

- The tag contains a microchip, which stores and processes the information, and an antenna, which receives and transmits the signal. It typically holds a unique serial number of the object.
- It operates in several frequency ranges: LF between 125 and 134 kHz, HF at 13.56 MHz, and UHF at 433 MHz and between 860 and 915 MHz.

RFID: Principle of Operation (1/2)

- The coils of the reader create a magnetic field when carrying electrical current.
- The antenna of the tag receives that excitation, which allows it to reply to the reader.
- In passive tags, the energy received in the excitation is used to power the operation of the circuit itself, doing away with a battery.

RFID: Principle of Operation (2/2)

- The tag replies with data, which may be a simple identifier or the answer to a challenge sent by the reader.
- The reader is normally connected to a host computer or to a network, where the collected information is processed.

Passive and Active RFID Tags (1/2)

- A passive tag holds no battery and uses the energy of the radio wave from the interrogator to return the stored information.
- Advantages of passive tags: they work without a battery, with a useful life of over twenty years; they have low manufacturing cost; and they are very small, with dimensions comparable to a grain of rice.

Passive and Active RFID Tags (2/2)

- Limitations of passive tags: they can only be read at very short distances, which restricts the possible applications, and they can hardly integrate sensors requiring electrical power.
- An active tag has a battery of its own, which gives it greater read range and allows additional functionality to be integrated, at the cost of higher price and periodic maintenance.

NFC: Near Field Communication (1/2)

- A set of communication protocols between electronic devices at very short distance, typically between 0 and 2 cm, reaching up to 10 cm.
- It is a subset of RFID technology, designed specifically for devices in close proximity.
- It uses inductive electromagnetic fields, unlike Bluetooth and Wi-Fi, which rest on radio transmission proper.

NFC: Near Field Communication (2/2)

- It offers a low throughput connection with simple setup, often used to initiate more capable wireless connections.
- It allows communication between two powered devices (both active) or between a powered device and a passive element with no power source of its own.

NFC: Technical Specifications (1/2)

- It operates at 13.56 MHz, in an ISM band available globally and unlicensed.
- It supports throughputs of 106, 212 and 424 kbit/s, over the air interface defined in standard ISO/IEC 18000-3.
- The theoretical operating distance with standardised compact antennas reaches 20 cm, but the practical distance is around 4 cm.
- It was approved as an ISO/IEC standard in December 2003 and later as an ECMA standard.

NFC: Technical Specifications (2/2)

- The reference standards are ISO/IEC 18092 (ECMA-340), which defines the NFCIP-1 interface and protocol, and ISO/IEC 21481 (ECMA-352), which defines NFCIP-2. The technology also incorporates standard ISO/IEC 14443.
- The NFC Forum defined a common data format, called NFC Data Exchange Format (NDEF).

NFC: Modes of Operation (1/2)

- **Reader/writer:** the active device reads or changes the content of a passive NFC tag.
- **Peer-to-peer:** two active devices exchange information with each other, a mode standardised by the NFC Forum in 2009.
- **Card emulation:** the device behaves towards the reader as a conventional contactless card, which underpins payment applications.
- An active device, typically a mobile phone with NFC, collects information from tags, exchanges data with other compatible devices and can change the content of the tag when authorised.

NFC: Modes of Operation (2/2)

- An NFC reader continuously emits radio frequency carrier signals and watches the received signals waiting for data.
- A passive tag has no power source of its own: it absorbs the energy emitted by the reader and replies with modulated information.

NFC: Tag Types and Applications (1/2)

- **Types 1 and 2:** based on standard ISO 14443-A, these are read and write tags, which the user may configure as read only. Available memory is 48 or 96 bytes, expandable to 2 KB.
- **Type 3:** based on the Japanese standard JIS X 6319-4, known as Sony FeliCa, with the read and write configuration set at manufacture.
- Current applications include smart posters, mobile payment, sharing of small files such as contacts, and peer to peer interaction scenarios.

NFC: Tag Types and Applications (2/2)

- The technology is also used in electronic identity documents and in access cards.
- **Advantages:** convenience of use, with a single tap; versatility, applicable from bank cards to transport passes and loyalty systems.
- **Limitations:** dependence on companies adopting the ecosystem, and exposure to security risks as mobile phones concentrate ever more sensitive information.

Bluetooth: General Overview (1/2)

- A low power wireless technology operating at short range in the 2.4 GHz ISM band.
- It emerged as a wireless alternative to cable connections, supporting the creation of personal area networks (PAN).
- It supports throughput of the order of 1 Mbit/s, suited to data traffic and, in more recent versions, to video traffic.
- It uses Frequency Hopping Spread Spectrum, which increases robustness against interference.

Bluetooth: General Overview (2/2)

- It operates in connected and non connected mode, according to whether an association between devices has been established.
- Power classes determine the range: class 1, with 100 mW (20 dBm), reaches about 100 m; class 2, with 2.5 mW (4 dBm), about 10 m; class 3, with 1 mW (0 dBm), about 1 m.
- Bluetooth 5 quadrupled the range compared with the previous generation, steering the technology towards IoT scenarios.

ZigBee: General Overview (1/2)

- A low cost, widely available technology aimed at sensor networks with reduced energy consumption.
- It defines three distinct roles for the devices making up the network.
- **Coordinator:** acts as the root of the network and as a bridge to other networks, being responsible for forming it.
- **Router:** an intermediate device that lets data cross the network towards other devices, extending coverage.

ZigBee: General Overview (2/2)

- **End device:** of limited functionality, it communicates only with its parent node, which lets it spend most of the time at rest and save energy.
- This organisation will be revisited in detail in the lesson dedicated to ZigBee.

Wi-Fi: Variants and Ranges (1/2)

- **IEEE 802.11a:** operates at 5 GHz, with throughput up to 54 Mbit/s and typical range of 10 m.
- **IEEE 802.11b:** operates at 2.4 GHz, with throughput up to 11 Mbit/s and typical range of 140 m.
- **IEEE 802.11g:** operates at 2.4 GHz, with throughput up to 54 Mbit/s and range of 140 m.
- **IEEE 802.11n:** operates at 2.4 GHz and 5 GHz, with throughput up to 450 Mbit/s and typical range of 250 m.

Wi-Fi: Variants and Ranges (2/2)

- **IEEE 802.11ah:** operates at 900 MHz, with throughput of the order of 8 Mbit/s, favouring range and consumption over throughput.
- The high consumption of conventional Wi-Fi variants limits their application in battery powered devices.

Wi-Fi HaLow (IEEE 802.11ah) (1/2)

- A Wi-Fi variant designed specifically for Internet of Things scenarios, operating below 1 GHz.
- Its reduced energy consumption puts it in direct competition with Bluetooth in sensing applications.
- It offers higher throughput and wider coverage than conventional short range technologies.

Wi-Fi HaLow (IEEE 802.11ah) (2/2)

- Operating at lower frequencies improves penetration through obstacles, which favours coverage inside buildings.
- It is an intermediate point between classical wireless local networks and the long range, low power technologies covered in the next lesson.

Trade-offs in Technology Selection (1/2)

- No technology combines low cost, long range and high bandwidth at once: the choice always means giving way on one of the axes.
- This principle is stated, ironically but accurately, in RFC 1925, known as “The Twelve Networking Truths”: of good, fast and cheap, you may pick only two.
- Short range technologies such as Bluetooth or ZigBee favour consumption and cost, sacrificing range.

Trade-offs in Technology Selection (2/2)

- Cellular technologies offer coverage and quality of service, but bring licensing costs and higher consumption.
- RFID and NFC offer very low cost per tag and need no power supply, at the price of a range of centimetres and the absence of a network proper.

Thank you.

Questions and comments